

# Queues and asynchronous work

Project: Grounded Learning Service

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Caching and stale information

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sandboxing and resource limits

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Queues and asynchronous work. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Queues and asynchronous work. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply a new case independently

Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

### My reasoning and evidence

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## 6. W2 Trace the worked case

Worker commits job J then crashes before queue acknowledgement. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

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## 7. W3 Audit the limit

A report omits this qualification: Queue ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

### My reasoning and evidence

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## 8. W4 Apply a new case independently

Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

### My reasoning and evidence

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

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## 10. Project checkpoint and reflection

Add a labelled queues and asynchronous work evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of queues and asynchronous work. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

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## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Queues and asynchronous work

Queued is not complete. A conversion service queues job J12 rather than processing it during submission.

States: queued, running, succeeded, failed. Submission returns J12 with queued status.

## Queues and asynchronous work

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Accept request

J12 acknowledges receipt, not completed conversion.



#### 2. Start worker

The worker marks J12 running before processing.



#### 3. Publish checked result

Only a checked output permits succeeded status.

### CHANGE ONE THING

The worker crashes before producing checked output.

Explain the changed result using the matching lesson.

Explain what the queued J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_

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**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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